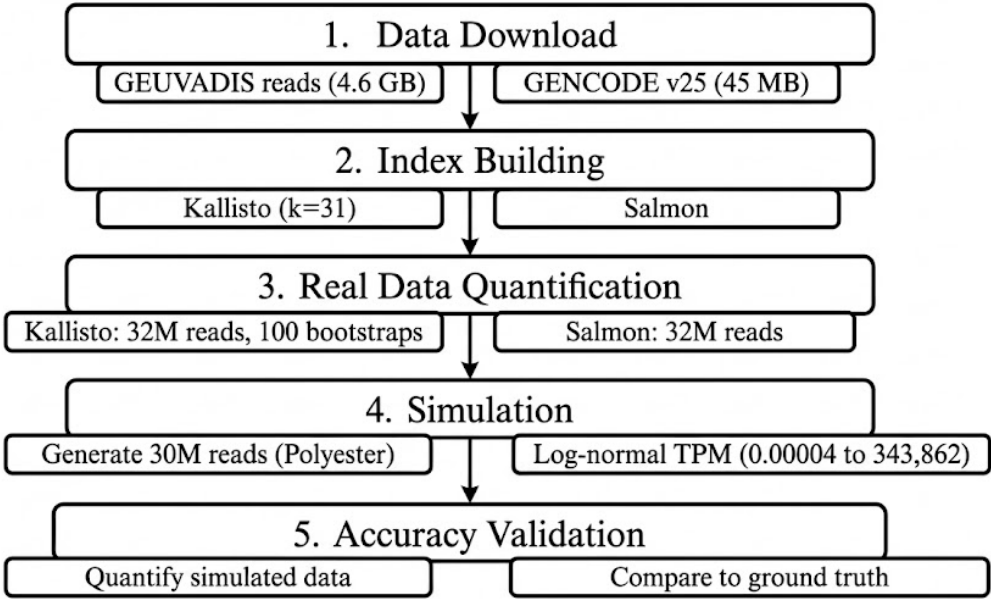


Reproducing Kallisto: Fast & Accurate RNA-seq Quantification via Pseudoalignment

SECTION: Background • RNA-seq quantification traditionally requires slow alignment (hours to days) • Kallisto (2016) introduced “pseudoalignment” - identifying transcript compatibility via k-mer matching without base-level alignment • Claims: <10 minute runtime with near-optimal accuracy ($\rho \approx 0.97-0.99$) • Uses transcriptome de Bruijn graph (T-DBG) with $k=31$
SECTION: Paper’s Key Claims • SPEED: Process 30M reads in <10 minutes on laptop • ACCURACY: Spearman $\rho \approx 0.97-0.99$ vs. ground truth (RSEM simulation) • UNCERTAINTY: Bootstrap resampling (100 reps) for confidence intervals • EFFICIENCY: ~3-4 GB RAM vs. 12+ GB for alignment-based tools
SECTION: Reproduction Methodology • Data (Same as Paper): – Sample: GEUVADIS NA12716 (ERR188021) – Transcriptome: GENCODE v25 (2016 release), Reads: 32.5M paired-end (75bp) – Simulation: 30M synthetic reads with log-normal TPM distribution • Tools: Kallisto v0.50 (100 bootstraps), Salmon v1.10 (comparison), Polyester (simulation, see limitations), Snakemake (automation) • Environment: Apple M1 MacBook Pro (Rosetta 2), Conda environment (kallisto-full)

SECTION: Pipeline Workflow



COMPARISON OF RESULTS

Original Paper (Bray et al. 2016)	My Reproduction
 • Speed: <10 min (30M reads)	 • Speed: 34 min (32M reads, 100 bootstraps)
 • Accuracy: $\rho = 0.97-0.99$	 • Accuracy: $\rho = 0.844$
 • Method: RSEM simulation	 • Method: Polyester simulation